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4 / 10 submitted

Q1. Reverse a list

— Not Submitted —

Q2. Reverse a string

— Not Submitted —

Q3. Searching

— Not Submitted —

Q4. Searching

CODE

```
#include <stdio.h>

int main() {
    int n, target, count = 0;

    if (scanf("%d", &n) != 1 || n ≤ 0) {
        return 0;
    }

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    scanf("%d", &target);

    for (int i = 0; i < n; i++) {
        if (arr[i] == target) {
            count++;
        }
    }

    printf("Occurrence = %d\n", count);

    return 0;
}
```

INPUT

```
6
1 2 5 2 3 2
2
```

OUTPUT

Occurrence = 3

Q5. Array**CODE**

```
#include <stdio.h>

int binarySearch(int arr[], int n, int target) {
    int low = 0;
    int high = n - 1;

    while (low ≤ high) {
        int mid = low + (high - low) / 2;

        if (arr[mid] == target) {
            return mid;
        }

        if (arr[mid] < target) {
            low = mid + 1;
        } else {
            high = mid - 1;
        }
    }

    return -1;
}

int main() {
    int n, target;

    if (scanf("%d", &n) ≠ 1 || n ≤ 0) {
        return 0;
    }

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    scanf("%d", &target);

    int result = binarySearch(arr, n, target);

    if (result ≠ -1) {
        printf("Element found at index %d\n", result);
    } else {
        printf("Element not found\n");
    }

    return 0;
}
```

INPUT

```
5
10 20 30 40 50
30
```

OUTPUT

```
Element found at index 2
```

Q6. Merging

— Not Submitted —

Q7. Largest and Smallest element

— Not Submitted —

Q8. Second Largest Element**CODE**

```
#include <stdio.h>
#include <limits.h>

int main() {
    int n;

    if (scanf("%d", &n) != 1 || n < 2) {
        return 0;
    }

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int largest = INT_MIN;
    int second_largest = INT_MIN;

    for (int i = 0; i < n; i++) {
        if (arr[i] > largest) {
            second_largest = largest;
            largest = arr[i];
        } else if (arr[i] > second_largest && arr[i] != largest) {
            second_largest = arr[i];
        }
    }

    if (second_largest == INT_MIN) {
        printf("No second largest element\n");
    } else {
        printf("Second Largest = %d\n", second_largest);
    }

    return 0;
}
```

INPUT

5
12 35 1 10 34

OUTPUT

Second Largest = 34

Q9. Occurrence of an elemnet**CODE**

```
#include <stdio.h>

int main() {
    int n, target;

    printf("");
    if (scanf("%d", &n) != 1 || n ≤ 0) {
        return 0;
    }

    int arr[n];
    printf("Enter %d elements:\n", n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    printf("");
    scanf("%d", &target);

    int first = -1, last = -1;

    for (int i = 0; i < n; i++) {
        if (arr[i] == target) {
            if (first == -1) {
                first = i;
            }
            last = i;
        }
    }

    if (first != -1) {
        printf("First Occurrence = %d\n", first);
        printf("Last Occurrence = %d\n", last);
    } else {
        printf("Element not found\n");
    }

    return 0;
}
```

INPUT

```
6
1 2 5 2 3 2
2
```

OUTPUT

```
Enter 6 elements:
First Occurrence = 1
Last Occurrence = 5
```

Q10. Missing Element

— Not Submitted —